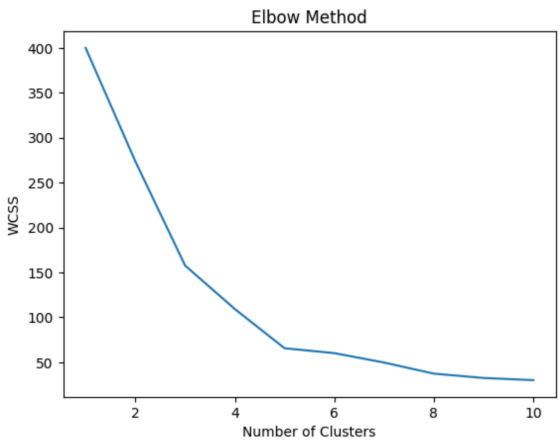
EXPERIMENT - 10

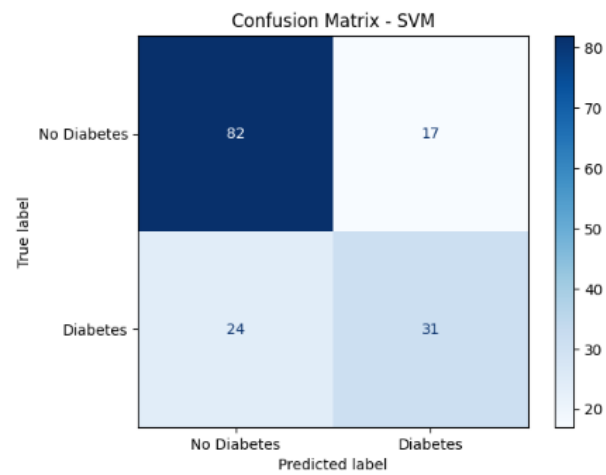
AIM: To apply **K-Means Clustering and Hierarchical Clustering** on a sample dataset to identify meaningful groups of customers based on their characteristics.

OUTPUT:

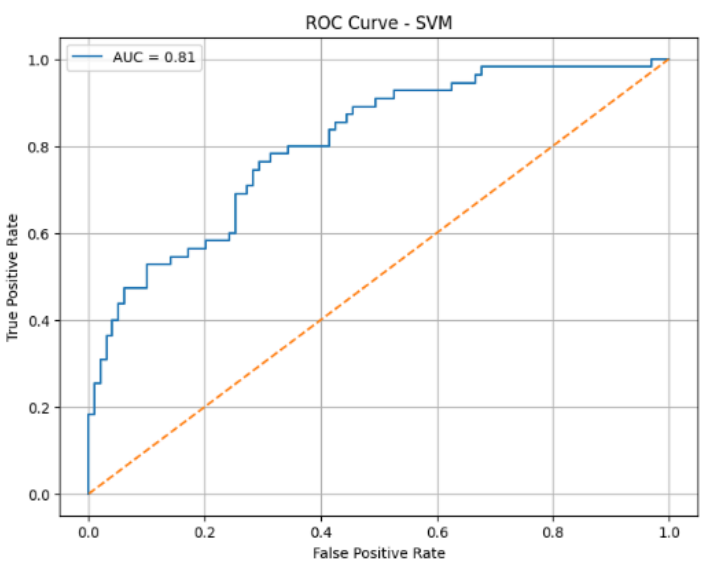
1. Elbow Method (Find Optimal K)



2. Apply and Visualize K-Means Clustering



3. ROC Curve




```
AUC Score: 0.8051423324150597
```

4. Hyparameter tuning

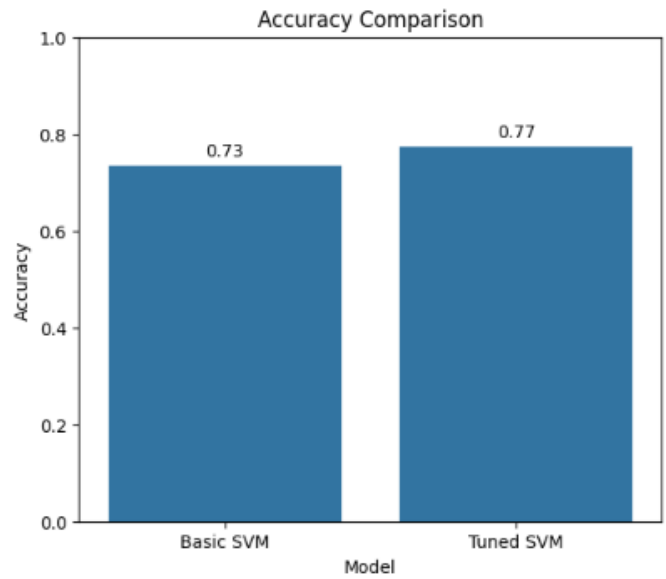
```
[CV] END .....................C=100, gamma=0.001, kernel=rbf; total time=   0.0s
[CV] END .....................C=100, gamma=0.001, kernel=rbf; total time=   0.1s
[CV] END .....................C=100, gamma=0.001, kernel=rbf; total time=   0.0s
[CV] END .....................C=100, gamma=0.001, kernel=rbf; total time=   0.0s
[CV] END .................C=100, gamma=0.001, kernel=linear; total time=   1.0s
[CV] END .................C=100, gamma=0.001, kernel=linear; total time=   1.1s
[CV] END .................C=100, gamma=0.001, kernel=linear; total time=   0.8s
[CV] END .................C=100, gamma=0.001, kernel=linear; total time=   1.0s
[CV] END .................C=100, gamma=0.001, kernel=linear; total time=   1.0s
Best Parameters: {'C': 100, 'gamma': 0.001, 'kernel': 'rbf'}
Best Cross Validation Score: 0.7720111955217913
```

5. Evaluate the best model

```
Optimized Accuracy: 0.7727272727272727

Classification Report:
              precision    recall  f1-score   support

           0       0.82      0.83      0.82        99
           1       0.69      0.67      0.68        55

    accuracy                           0.77       154
   macro avg       0.75      0.75      0.75       154
weighted avg       0.77      0.77      0.77       154
```

6. Comparison before vs After tuning